REMARKS ON SPLITTING CONICS OF SMOOTH PLANE SEXTICS

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Abstract.

1. Introduction

We work over the complex number field $\mathbb{C}$.

Let B be a smooth plane sextic. A smooth conic C is said to be a *splitting conic* of B if the local intersection multiplicity of B and C at every point of $B \cap C$ is even. A *splitting-conic configuration* is a pair (B, Γ) of a smooth plane sextic B and a non-ordered collection $\Gamma = \{C_1, \ldots, C_n\}$ of splitting conics of B. The number n of the conics in Γ is the *size* of the configuration.

We investigate splitting configurations of smooth plane sextics. Our main results are the following.

(A) We exhibit a splitting-conic configuration of size 120, which is associated with the del Pezzo surface of degree 1 and the root lattice of type E_8.

(B) We give the complete classification of splitting-conic configurations of size ≤ 6 up to deformation equivalence.

(C) We construct examples of splitting-conic configurations of large sizes by means of the Leech lattice. The maximum size we have found is 4455.

As the phrase "splitting" suggests, the main idea of our investigation is to consider the double covering

$$\pi_B \colon X_B \to \mathbb{P}^2$$

branched along the smooth plane sextic B. The double plane X_B is a K3 surface, and the inverse image $\pi_B^{-1}(C)$ of a splitting conic C of B splits into two (-2)-curves C^+ and C^-, which we call the lifts of C. For disctinct splitting conics C_i and C_j of B, we have

$$\langle C_i^+, C_j^+ \rangle \in \{0, 1, 2, 3, 4\}, \qquad \langle C_i^+, C_j^+ \rangle + \langle C_i^-, C_j^+ \rangle = 4.$$

To describe the intersection patterns of the lifts of splitting conics, we introduce an invariant $\gamma(B, \Gamma)$. Let $\mathcal{G}_n$ be the set of weighted complete graphs with the set of vertices $\{1, \ldots, n\}$, and each edge ij $(i \neq j)$ is weighted by an integer $w_{ij} \in \{0, 1, 2, 3, 4\}$. The group

$$G_n = \mathfrak{S}_n \ltimes (\mathbb{Z}/2\mathbb{Z})^n$$

acts on $\mathcal{G}_n$, where the symmetric group $\mathfrak{S}_n$ acts on $\mathcal{G}_n$ via permutations of labeling of the vertices, and the i-th factor $\mathbb{Z}/2\mathbb{Z}$ of $(\mathbb{Z}/2\mathbb{Z})^n$ acts on $\mathcal{G}_n$ by changing the weights w_{ij} of the edges containing the vertex i to $4 - w_{ij}$.

Definition 1.1. Let (B, Γ) be a splitting-conic configuration of size n. We label the members of Γ as $C_1, \ldots, C_n$, and choose a lift C_i^+ of each member C_i. Then we obtain a weighted graph $g(B, \Gamma) \in \mathcal{G}_n$ by putting the weight w_{ij} to be $\langle C_i^+, C_j^+ \rangle$. Let $\gamma(B, \Gamma) \in \mathcal{G}_n/G_n$ be the G_n-orbit of the graph $g(B, \Gamma)$. Then it is obvious that $\gamma(B, \Gamma)$ does not depend on the choice of the labelling of Γ and the choices of the lifts C_i^+.

1.1. **Splitting-conic configuration** (B_ι, Δ_{120})**.**

Definition 1.2. A splitting-conic configuration (B, Γ) of size n is said to be of general singularity type if each member C_i of $\Gamma = \{C_1, \ldots, C_n\}$ intersects B at distinct six points and no three curves in $B, C_1, \ldots, C_n$ have a common point, or equivalently, the singulatieis of the union of $B, C_1, \ldots, C_n$ consists of $6n$ tacnodes and $n(n-1)/2$ ordinary nodes.

From the theory of del Pezzo surfaces [**?**], we obtain the following simple observation:

Proposition 1.3. *For $\nu = 0, 2, 4, 6$, let $f_\nu(x_0, x_1)$ be a general homogeneous polynomial of degree ν. Then the smooth plane sextic B_ι defined by the equation*

$$f_6(x_0, x_1) + f_4(x_0, x_1)x_2^2 + f_2(x_0, x_1)x_2^4 + f_0(x_0, x_1)x_2^6 = 0 \tag{1.1}$$

possesses exactly 120 *splitting conics. Let Δ_{120} be the union of these conics. Then (B_ι, Δ_{120}) is of general singularity type.*

The invariant $\gamma(B_\iota, \Delta_{120})$ is calculated by the following. Let $\mathbb{E}_8$ denote the positive definite root lattice of type E_8 with the intersection form $(\ ,\ )_E$, and let $(\mathbb{E}_8)_2$ denote the set of vectors r of norm $(r, r)_E = 2$.

Proposition 1.4. *Let $\widetilde{\Delta}_{120}$ be the set of lifts of splitting conics of B_ι, on which the switching $C^+ \leftrightarrow C^-$ acts. Then there exists a bijection*

$$\rho\colon \widetilde{\Delta}_{120} \xrightarrow{\sim} (\mathbb{E}_8)_2$$

such that

(i) *the switching on $\widetilde{\Delta}_{120}$ corresponds to the involution $r \leftrightarrow -r$ on $(\mathbb{E}_8)_2$, and*
(ii) *for $\widetilde{C}, \widetilde{C}' \in \widetilde{\Delta}_{120}$, we have*

$$\langle \widetilde{C}, \widetilde{C}' \rangle \ = \ -2\,(\,\rho(\widetilde{C}), \rho(\widetilde{C}')\,)_E \ + \ 2.$$

We construct examples of Zariski multiples from the configuration (B_ι, Δ_{120}).

Definition 1.5. Let D and D' be reduced, possibly reducible plane curves. We say that D and D' are of the same combinatorial type if there exist tubular neighborhoods T of D and T' of D' in $\mathbb{P}^2$ such that (T, D) and (T', D') are homeomorphic, whereas we say that D and D' are of the same topological type if $(\mathbb{P}^2, D)$ and $(\mathbb{P}^2, D')$ are homeomorphic.

For a subset Δ of Δ_{120}, let D_Δ denote the union of B_ι and the conics in Δ. Note that the combinatorial type of D_Δ depends only on the size of Δ.

We put

$$\mathcal{S}_n := \left\{ \{r_1, \dots, r_n\} \subset (\mathbb{E}_8)_2 \;\middle|\; \begin{array}{l} r_1, \dots, r_n \in \mathbb{E}_8 \text{ are distinct vectors of norm } 2 \\ \text{that span } \mathbb{E}_8 \otimes \mathbb{Q} \text{ over } \mathbb{Q} \end{array} \right\},$$

on which the Weyl group $W(\mathbb{E}_8)$ acts.

Theorem 1.6. *The set of plane curves D_Δ, where Δ runs through the subsets of Δ_{120} of size $|\Delta| = n$, are divided into at least s_n distinct topological types, where s_n is the number of $W(\mathbb{E}_8)$-orbits in $\mathcal{S}_n$.*

See Remark **??** for a rough estimate of the number s_n. Thus we obtain examples of Zariski N-ples for large N by choosing various subsets Δ of Δ_{120}.

1.2. **Classification of splitting-conic configurations of small size.** The splitting conic configuration (B_ι, Δ_{120}) above has the property that every pair of lifts of splitting conics has even intersection numbers. There exist, of course, splitting conic configurations such that some pairs of lifts have odd intersection numbers.

Definition 1.7. We say that a splitting conic configuration (B, Γ) is *full* if there exist no splitting conics of B other than the members of Γ.

For a splitting conic configuration (B, Γ), we denote by S_B the Picard lattice of the K3 surface X_B, by $h \in S_B$ the class of the pullback of a line on $\mathbb{P}^2$, by $\widetilde{\Gamma} \subset S_B$ the set of classes of lifts of the splitting conics in Γ, and by $S_{(B,\Gamma)}$ the submodule of S_B generated by h and $\widetilde{\Gamma}$.

Definition 1.8. We say that a splitting conic configuration (B, Γ) is *saturated* if $S_{(B,\Gamma)}$ is of finite index in S_B.

Definition 1.9. Two splitting configurations (B, Γ) and (B', Γ') are said to be *deformation equivalent* if there exists a continuous family of splitting conic configurations (B_t, Γ_t) parameterized by $t \in [0, 1]$ such that $(B_0, \Gamma_0) = (B, \Gamma)$ and $(B_1, \Gamma_1) = (B', \Gamma')$.

Our goal is to classify all full splitting conic configurations up to deformation equivalence. It is easy to see that any splitting conic configuration is deformation equivalent to a saturated splitting conic configuration. Hence it is enough to classify all full and saturated splitting conic configurations. We carried out this task for the cases $\operatorname{rank} S_B \leq 7$.

$\operatorname{rank} S_B$	number
3	3
4	9
5	45
6	326
7	5192

TABLE 1.1. Number of deformation equivalence classes

Theorem 1.10. *The number of deformation equivalence classes of full and saturated splitting conic configurations* (B, Γ) *with* $\operatorname{rank} S_B$ *is indicated in Table 1.1.*

For the deformation classes with small $\operatorname{rank} S_B$, we present more detailed data in Table 1.2. In this table, r is $\operatorname{rank} S_B = \operatorname{rank} S_{(B,\Gamma)}$, d_B is the discriminant of the lattice D_B, e is the index of $S_{(B,\Gamma)}$ in S_B, $|\Gamma|$ is the number n of splitting conics, Λ is the number of splitting lines, and γ is the vector $[k_{04}, k_{22}, k_{13}]$ describing the G_n-orbit $\gamma(B, \Gamma)$ of the weighted complete graphs, where k_{04} (resp. k_{22}, resp. k_{13}) is the number of edges with weight 0 or 4 (resp. 2, resp. 1 or 3).

No.	r	d	e	$\lvert\Gamma\rvert$	$\lvert\Lambda\rvert$	γ
1	3	24	1	3	0	$[3,0,0]$
2	3	30	1	2	0	$[0,0,1]$
3	3	32	1	2	0	$[0,1,0]$
4	4	64	1	6	0	$[12,3,0]$
5	4	72	1	5	0	$[6,0,4]$
6	4	88	1	4	0	$[3,1,2]$
7	4	96	1	4	0	$[3,3,0]$
8	4	100	1	3	0	$[0,0,3]$
9	4	108	1	3	0	$[0,0,3]$
10	4	112	1	3	0	$[0,1,2]$
11	4	120	1	3	0	$[0,2,1]$
12	4	128	1	3	0	$[0,3,0]$
13	5	128	1	12	0	$[48,18,0]$
14	5	160	1	10	0	$[30,15,0]$
15	5	162	1	9	0	$[18,0,18]$
16	5	168	1	9	0	$[21,6,9]$
17	5	192	1	8	0	$[15,5,8]$
18	5	210	1	7	0	$[9,2,10]$
19	5	216	1	7	0	$[9,0,12]$
20	5	224	1	7	0	$[12,5,4]$
21	5	232	1	7	0	$[12,6,3]$
22	5	240	1	6	0	$[6,1,8]$
23	5	242	1	6	0	$[6,3,6]$
24	5	250	1	5	0	$[0,0,10]$
25	5	256	1	6	0	$[6,5,4]$
26	5	256	1	7	0	$[12,9,0]$
27	5	258	1	6	0	$[6,2,7]$
28	5	264	1	6	0	$[6,3,6]$
29	5	280	1	5	0	$[3,2,5]$
30	5	288	1	6	0	$[6,5,4]$
31	5	288	1	6	0	$[6,9,0]$
32	5	290	1	5	0	$[3,2,5]$
33	5	306	1	5	0	$[3,2,5]$
34	5	312	1	5	0	$[3,2,5]$
35	5	320	1	4	0	$[0,1,5]$
36	5	320	1	5	0	$[3,3,4]$
37	5	322	1	5	0	$[3,3,4]$
38	5	328	1	5	0	$[3,4,3]$
39	5	330	1	4	0	$[0,0,6]$
40	5	352	1	5	0	$[3,5,2]$
41	5	360	1	4	0	$[0,1,5]$
42	5	360	1	5	0	$[3,6,1]$
43	5	370	1	4	0	$[0,2,4]$
44	5	378	1	4	0	$[0,0,6]$
45	5	384	1	4	0	$[0,1,5]$
46	5	384	1	4	0	$[0,2,4]$
47	5	384	1	5	0	$[3,7,0]$
48	5	392	1	4	0	$[0,2,4]$
49	5	400	1	4	0	$[0,3,3]$
50	5	402	1	4	0	$[0,2,4]$
51	5	416	1	4	0	$[0,3,3]$
52	5	418	1	4	0	$[0,3,3]$
53	5	432	1	4	0	$[0,3,3]$
54	5	448	1	4	0	$[0,4,2]$
55	5	450	1	4	0	$[0,4,2]$
56	5	480	1	4	0	$[0,5,1]$
57	5	512	1	4	0	$[0,6,0]$

TABLE 1.2. Splitting conic configurations

See Section **??** for the proof of this proposition and further geometric properties of this configuration of 120 conics.

Let B be a smooth plane sextic. A smooth conic C is called a splitting conic to B if the local intersection multiplicity of B and C at every point of $B \cap C$ is even. Let $\pi_B \colon X_B \to \mathbb{P}^2$ be the double cover branched along B. Then X_B is a K3 surface. A smooth rational curve on X_B is called a (-2)-curve. Let C be a splitting conic of B. Then the inverse image $\pi_B^{-1}(C)$ is the union of two smooth rational curves $\widetilde{C}^+$ and $\widetilde{C}^-$. Let C_1 and C_2 be two distinct splitting conics of B. Then we have

$$\langle \widetilde{C}_1^+, \widetilde{C}_2^+ \rangle \in \{0, 1, 2, 3, 4\}, \qquad \langle \widetilde{C}_1^+, \widetilde{C}_2^+ \rangle + \langle \widetilde{C}_1^+, \widetilde{C}_2^- \rangle = 4.$$

Let k be an integer with $k \geq 2$. Let E_k be the set of two-element subsets of $\{1, \ldots, k\}$, and let W_k denote the set of mappings

$$w \colon E_k \to \{0, 1, 2, 3, 4\}.$$

Thus W_k is the set of weighted complete graphs on k vertices $\{1, \ldots, k\}$ with weights $0, 1, 2, 3, 4$. Then the symmetric group $\mathfrak{S}_k$ acts on W_k via its natural action on the vertex set $\{1, \ldots, k\}$:

$$^{\tau}w(\{i, j\}) := w(\{i^{\tau}, j^{\tau}\}).$$

We let the multiplicative group $\{\pm 1\}$ act on $\{0, 1, 2, 3, 4\}$ by

$$a^s := \begin{cases} a & \text{if } s = 1, \\ 4 - a & \text{if } s = -1. \end{cases}$$

Then $\{\pm 1\}^k$ acts on W_k by

$$^{(s_1, \ldots, s_k)}w(\{i, j\}) = (w(\{i, j\}))^{s_i s_j}.$$

Thus the semidirect product $G := \{\pm 1\}^k \rtimes \mathfrak{S}_k$ acts on W_k. We call a G-orbit in W_k a splitting graph.

Remark 1.11. The group $\mathfrak{S}_k$ acts on $\{\pm 1\}^k$ by

$$(^{\tau}s)_i := s_{i^{\tau}}.$$

The product in G is given by

$$(s, \tau)(s', \tau') = (s\ ^{\tau}s', \tau\tau').$$

The action of G on W_k is given by

$$^{(s,\tau)}w = {}^{s}(^{\tau}w).$$

Remark 1.12. Our construction is a variant of Shirane's splitting graph adapted to collections of splitting conics.

Suppose that a smooth plane sextic B has a collection Γ of k distinct splitting conics. By labelling these splitting conics as $C_1, \dots, C_k$, and choosing an irreducible component $\widetilde{C}_i^+$ from a the two irreducible components of $\pi_B^{-1}(C_i)$ for each $i = 1, \dots, k$, we obtain an element w_Γ of W_k by

$$w_\Gamma(\{i,j\}) := \langle \widetilde{C}_i^+, \widetilde{C}_j^+ \rangle.$$

Note that the G-orbit $[w_\Gamma] := G \cdot w_\Gamma$ is independent of the labelling and of the choices of the irreducible components $\widetilde{C}_i^+$.

Theorem 1.13. *Let γ be a splitting graph in the first column of Table* **??**. *Then the family $\mathcal{F}_\gamma \subset \mathbb{P}_*(\Gamma(\mathbb{P}^2, \mathcal{O}(6)))$ of smooth plane sextics B possessing a collection Γ of splitting conics such that $[w_\Gamma] = \gamma$ is irreducible, and a general member of $\mathcal{F}_\gamma$ has a collection $\Gamma' \supset \Gamma$ of splitting conics such that $[w_{\Gamma'}] = \gamma'$, where γ' is the splitting graph in the same row of Table* **??** *as γ.*

For example, the first row of Table **??** has the following interpretation. Let B be a general plane sextic in the family of smooth plane sextics possessing two splitting conics C_1 and C_2 such that $\langle \widetilde{C}_1^+, \widetilde{C}_2^+ \rangle = 0$ for a suitable choice of the lifts $\widetilde{C}_1^+, \widetilde{C}_2^+$. Then B has another splitting conic C_3 such that

$$\langle \widetilde{C}_1^+, \widetilde{C}_3^+ \rangle = \langle \widetilde{C}_2^+, \widetilde{C}_3^+ \rangle = 0$$

for a suitable choice of $\widetilde{C}_3^+$.

1.3. **Enumeration of splitting conics.** Let B be a smooth plane sextic, let $\pi_B \colon X_B \to \mathbb{P}^2$ be the double cover branched along B, and let $j \colon X_B \to X_B$ be the involution with respect to the double cover $\pi_B \colon X_B \to \mathbb{P}^2$. We denote by $h \in \mathrm{Pic}(X_B)$ the class of the pull-back of a line of $\mathbb{P}^2$. Then h is ample. The involution j fixes h and acts as multiplication by -1 on the orthogonal complement $h^\perp$ in $\mathrm{Pic}(X_B)$. The set $\mathcal{R}_d$ of classes of smooth rational curves C of degree $\langle C, h \rangle = d$ on X_B is enumerated inductively on d as follows:

$$\mathcal{R}_d := \left\{ r \in \mathrm{Pic}(X_B) \;\middle|\; \begin{array}{l} \langle r, r \rangle = -2,\ \langle r, h \rangle = d, \\ \text{for all } d' < d \text{ and for all } r' \in \mathcal{R}_{d'}, \text{ we have } \langle r, r' \rangle \geq 0 \end{array} \right\}.$$

Then the set of classes of splitting conics of B is equal to the set of j-orbits in $\mathcal{R}_2$.

1.4. **Proof of Proposition 1.3.** Throughout this subsection, let B denote a smooth plane sextic defined by an equation of the form (1.1). Let $\iota \colon \mathbb{P}^2 \to \mathbb{P}^2$ be the involution given by

$$[x_0 : x_1 : x_2] \mapsto [x_0 : x_1 : -x_2].$$

Then B is ι-invariant. We show that the Picard lattice $\mathrm{Pic}(X_B)$ of the $K3$ surface X_B is of rank 9 and admits a basis such that

(a) the Gram matrix of $\mathrm{Pic}(X_B)$ is the diagonal matrix

$$\mathrm{diag}(2, -2, \dots, -2),$$

(b) the class $h \in \mathrm{Pic}(X_B)$ of the pull-back of a line of $\mathbb{P}^2$ is written as

$$h = (3, -1, \ldots, -1).$$

We denote the space of all plane sextics by

$$\mathcal{P}_6 := \mathbb{P}_*(\Gamma(\mathbb{P}^2, \mathcal{O}(6))),$$

and let $\mathcal{S} \subset \mathcal{P}_6$ be the Zariski open subset of smooth plane sextics. Then the period map gives an open immersion of $\mathcal{S}/\mathrm{PGL}(3,\mathbb{C})$ into a 19-dimensional period domain. The space $\mathcal{I}_\iota$ of ι-invariant sextics is a 15-dimensional subspace of $\mathcal{P}_6$. Since the centralizer of ι in $\mathrm{PGL}(3,\mathbb{C})$ is of dimension 4, the image of

$$\mathcal{I}_\iota \cap \mathcal{S} \quad \hookrightarrow \quad \mathcal{S} \quad \to \quad \mathcal{S}/\mathrm{PGL}(3,\mathbb{C})$$

is $15 - 4 = 11$. (Here we use the fact that a smooth plane sextic is of genus 10 and hence its automorphism group is finite.) Since B is a general point of $\mathcal{I}_\iota \cap \mathcal{S}$, the rank of $\mathrm{Pic}(X_B)$ is $20 - 11 = 9$. Let $\phi\colon \mathbb{P}^2 \to Q$ be the quotient morphism by the involution ι, where $Q \subset \mathbb{P}^3$ is the quadric surface of rank 3 defined by $z_0 z_2 - z_1^2 = 0$, and ϕ is given by

$$[x_0 : x_1 : x_2] \mapsto [z_0 : z_1 : z_2 : z_3] = [x_0^2 : x_0 x_1 : x_1^2 : x_2^2].$$

The fixed point $[0 : 0 : 1]$ of ι is mapped to the vertex A of Q and the fixed line $\{x_2 = 0\}$ is mapped to the plane section $Q \cap H_0$ of Q, where $H_0 = \{z_3 = 0\}$. The quotient morphism ϕ induces a double covering $B \to B_Q$ of a $(2,3)$-complete intersection $B_Q \subset Q$. Let $\varpi_B\colon Y_B \to Q$ be the double covering branched along B_Q and at the vertex A. Then Y_B is a del Pezzo surface of degree 1, and the morphism $\varpi_B\colon Y_B \to Q$ is the bi-anticanonical model of Y_B. We denote by α_W the anti-canonical class of Y_B. The morphism ϕ lifts to a double covering $\varphi\colon X_B \to Y_B$. Thus we obtain the following commutative diagram:

$$\begin{array}{ccc} X_B & \xrightarrow{\varphi} & Y_B \\ {\scriptstyle \pi_B}\downarrow & & \downarrow{\scriptstyle \varpi_B} \\ \mathbb{P}^2 & \underset{\phi}{\to} & Q. \end{array}$$

The branch locus of the double covering $\varphi\colon X_B \to Y_B$ is the pull-back $\varpi_B^{-1}(Q \cap H_0)$, which is a member of the bi-anticanonical system $|2\alpha_W|$. Since the pull-back by ϕ of a plane section $Q \cap H$ of Q is a conic on $\mathbb{P}^2$, we have $2h = \varphi^*(2\alpha_W)$. Since $\mathrm{Pic}(X_B)$ is torsion free, we obtain

$$h = \varphi^* \alpha_W.$$

The del Pezzo surface Y_B is obtained by blowing up a projective plane $\mathbf{P}^2$ at eight distinct points in general position. Hence $\mathrm{Pic}(Y_B)$ is isomorphic to $\mathrm{I}_{1,8}$, and has a basis $l, e_1, \ldots, e_8$, where l is the class of the pull-back of a line of $\mathbf{P}^2$ and $e_1, \ldots, e_8$ are the classes of exceptional divisors. We have

$$\alpha_W = 3l - e_1 - \cdots - e_8.$$

$\dim_{\mathbb{F}_2} M/\mathrm{I}_{1,8}(2)$	number	type of $R(h^\perp)$
0	1	0
1	135	$8A_1$
2	1575	$2D_4$
3	2025	D_8
4	270	E_8

TABLE 1.3. Overlattices M of $\mathrm{I}_{1,8}(2)$

Now we have

$$\mathrm{I}_{1,8}(2) \cong \varphi^*\mathrm{Pic}(Y_B) \quad \subset \quad \mathrm{Pic}(X_B).$$

Since $\mathrm{Pic}(X_B)$ is of rank 9, $\mathrm{Pic}(X_B)$ is an overlattice of $\mathrm{I}_{1,8}(2)$. The discriminant form of $\mathrm{I}_{1,8}(2)$ has 135 non-zero isotropic vectors. By Nikulin's theory of discriminant forms [**?**], we see that $\mathrm{I}_{1,8}(2)$ has $1+135+1575+2025+270$ even overlattices as in Table 1.3. Note that

$$h = \varphi^*(3l - e_1 - \cdots - e_8)$$

is ample, and hence

$$R(h^\perp) := \{\, r \in \mathrm{Pic}(X_B) \mid \langle r, r\rangle = -2, \ \ \langle r, h\rangle = 0 \,\}$$

should be empty. On the other hand, as is shown in Table 1.3, if the extension index $\mathrm{Pic}(X_B)/\mathrm{I}_{1,8}(2)$ is > 1, then $R(h^\perp)$ would have an element. Therefore the overlattice $\mathrm{Pic}(X_B)$ is in fact equal to $\varphi^*\mathrm{Pic}(Y_B)$ and hence has a basis $\varphi^*l, \varphi^*e_1, \ldots, \varphi^*e_8$ with the desired properties (a), (b).

Then we see that X_B has exactly 120 splitting conics by the method given in Section 1.3. Indeed, there are no vectors r in $\mathrm{Pic}(X_B)$ with $\langle r, r\rangle = -2$ and $\langle h, r\rangle = 1$, and there exists exactly 240 vectors r with $\langle r, r\rangle = -2$ and $\langle h, r\rangle = 2$. Therefore these 240 vectors are the classes of smooth rational curves. Moreover the involution j of the double covering $X_B \to \mathbb{P}^2$ acts on these 240 vectors without fixed points, and hence the 120 orbits under the involution j are in one-to-one correspondence with splitting conics.

In fact, these 240 vectors r are the pull-back of the 240 lines of the del Pezzo surface Y_B. Note that the mapping $B \mapsto B_Q$ gives an isomorphism between the space the projective space $\mathcal{I}_\iota$ of ι-invariant sextics and the linear system $|\mathcal{O}_Q(3)|$. Hence Y_B is a general member of the family $\mathcal{U}$ of bi-anticanonical models of del Pezzo surfaces of degree 1 studied in [**?**]. By [**?**, Proposition 4.4], the union of the 240 lines of Y_B has only ordinary double points, which implies that the 120 splitting conics of B are in fact six-tangential conics.

Remark 1.14. In fact, the automorphism group of the $K3$ surface X_B is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and the 240 curves obtained above are all the smooth rational curves on X_B.

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